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AISI 431 | 1.4057 | BS 431S29 | X17CrNi16-2 | Martensitic steel

Data sheet according to DIN EN 10088 for material number 1.4057

The material AISI 431, which equals 1.4057, BS 431S29, and X17CrNi16-2, is a stainless martensitic chromium-nickel steel with a chromium content of 15-17 %. Due to the high chromium content, this stainless steel has better corrosion resistance and toughness than comparable steels with a lower chromium content. The material AISI 431 (1.4057, BS 431S29, X17CrNi16-2) is magnetic (ferromagnetic) and has a tensile strength of 800-950 N/mm² (in heat treatment condition +QT800) with a density of 7.0 kg/dm³ (at room temperature). Due to its high strength and resistance, this steel is used, among other things, for the manufacture of machine parts subject to high mechanical stress, such as piston rods. The material can be used in the temperature range from -40°C to 400°C. It has good polishability, but only medium forgeability and poor machinability. Cold forming is possible. This stainless steel grade can be welded well with most welding processes.

The data sheet provides technical specifications for AISI 431 (1.4057) according to DIN EN 10088. It includes chemical composition, mechanical properties, and physical properties.

Chemical composition (mass fraction in % according to DIN EN 10088)	
C	0.05
Mn	0.05
P	0.015
S	0.005
Si	0.05
Cr	16.0 - 18.0
Ni	1.0 - 1.5

Mechanical properties at room temperature in % according to DIN EN 10088	
Yield strength (R _{p0.2})	800
Tensile strength (R _m)	950
Elongation (A ₅)	10

Mechanical properties at room temperature in % according to DIN EN 10088	
Impact strength (K _{IC})	20
Hardness (HRC)	28

[Stainless steel wire AISI 431 / 1.4057](#)
[Stainless steel bars AISI 431 / 1.4057](#)

Specifications of the material

EN-Grade: 1.4057
 EN-short name: X17CrNi16-2
 EN-standard: 10088
 Microstructure: martensit

Comparable standards and designations

AFNOR Z15CN16-02
 AISI 431
 Alloy 431
 AMS 5682
 B.S. 431S29
 ČSN 17145
 JIS SUS431
 PN 2H17N2
 RVS 431
 UNE F.3427
 UNS S43100
 SAE 431
 SS 2321



Chemical composition and properties of stainless steel AISI 431 (1.4057 / BS 431S29)

C	Si	Mn	P	S	N	Cr	Cu	Mo	Ni	Ti
0,12 - 0,22	≤ 1,00	≤ 1,50	≤ 0,04	≤ 0,03	-	15,0 - 17,0	-	-	1,50 - 2,50	-

Mass fraction in % according to DIN EN 10088-3

Abbreviations: **C** = carbon, **Cr** = chromium, **Cu** = copper, **Mn** = manganese, **Mo** = molybdenum, **N** = nitrogen, **Ni** = nickel, **P** = phosphorus, **S** = sulphur, **Si** = silicon, **Ti** = titanium

Stainless steel overview

Physical properties

Magnetizability: present

Density (kg/dm³): 7.0

Thermal conductivity (at up to 20°C): 25

Electronic resistance at room temperature (in Ω mm²/m): 0.70

Mechanical properties

Heat treatment condition annealed (+A)

Hardness in HB: 295

Tensile strength Rm in Mpa: Max. 950

Heat treatment condition quenched and tempered (+QT800) ≤ 60 and 60 < t ≤ 160

0.2 % yield strength (Rp0.2 in Mpa): 600

Tensile strength Rm in Mpa: 800-950

Elongation at break A in % (longitudinal)

≤ 60: 14

60 < t ≤ 160: 12

Heat treatment condition quenched and tempered (+QT900) ≤ 60 and 60 < t ≤ 160

0.2 % yield strength (Rp0.2 in Mpa): 700

Tensile strength Rm in Mpa: 900-1050

Elongation at break A in % (longitudinal)

≤ 60: 12

60 < t ≤ 160: 10

Minimum values of the 0.2 % proof stress at elevated temperatures

Temp in °C	100 °C	150 °C	200 °C	250 °C	300 °C	350 °C	400 °C
+QT900	565	525	505	490	470	430	375
+QT800	515	495	475	460	440	405	355

Weldability of stainless steel AISI 431 / 1.4057

CEV calculator

The material AISI 431 (1.4057, BS 431S29, X17CrNi16-2) is suitable for some welding processes, but certain precautions must be observed. Gas containing hydrogen or nitrogen must not be used when welding this material. If filler metal is required, materials 1.4430 (AISI AWS316L) or 1.4370 should be used. This applies in particular if the weld seam does not have high strength. Preheating can be omitted when using the filler materials. Otherwise,



preheating to 100°C-300°C is required. The material must not cool below 200°C during welding. If no additional post-treatment of the weld seam is carried out, it should also be noted that the mechanical-technical values of the material can vary greatly in relation to those of the base material.

Forgeability of the material

The martensitic chromium-nickel steel AISI 431 (1.4057, BS 431S29, X17CrNi16-2) has medium forgeability. During the forging process, the material should first be heated slowly to 850 °C. This should be followed by rapid heating. It should then be heated quickly to around 1180 °C. Forging can be carried out in the temperature range between around 1150 °C and 950 °C. The material should then cool down slowly. This material is not normally used for open-die forging and closed-die forging.

Heat treatment and hot forming

Hot forming: 800-1100 °C | Type of cooling: slow cooling

Annealing: 680-800 °C | Cooling type: furnace, air

Corrosion resistance of the material

As a chromium-nickel steel, the material AISI 431 (1.4057, BS 431S29, X17CrNi16-2) has good corrosion resistance. However, the material is susceptible to intergranular corrosion. Maximum corrosion resistance is achieved with a bright metallic surface.

Applications for stainless steel AISI 431 / 1.4057

Due to its high strength, stainless steel AISI 431 (1.4057, BS 431S29, X17CrNi16-2) is often used where parts are subject to high stresses. This includes mechanical engineering and turbine construction, the automotive industry and the aviation industry. The material is also used in the chemical, petrochemical and oil industries and in power plant construction.

Processing options

The processing options for this grade of stainless steel include cold forming and machining. The material also has good polishability. This material is not normally used for cold heading.

Please note

The information given in this data sheet has been compiled to the best of our knowledge and is based on the current version of the relevant standard. It is considered for reference only and we assume no liability for any errors.



Do you have any questions about our products or are you looking for a different stainless steel wire?

Get in touch with our team



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